QuantFolio — Project Summary

QuantFolio is a Python/Streamlit portfolio analytics terminal built for Indian markets (NSE + BSE) that also works with any Yahoo Finance symbol. It turns a list of holdings into professional-grade return, risk, optimisation, tax and machine-learning forecast analysis. Current release: v3.0.1, MIT-licensed.

Portfolio Input

Three routes, all normalised to `ticker, quantity, buy_price`:

- **Search** across ~7,570 bundled NSE/BSE instruments, offline, ranked by exchange turnover.
- **Broker CSV import** with automatic column detection, symbol resolution (RELIANCE → RELIANCE.NS) and a report of every rejected row with its reason.
- **ISIN-only import** with ISO 6166 check-digit validation and 5,175 bundled ISIN mappings.

An optional trade ledger (date, action, quantity, price, fees) unlocks FIFO tax lots, realised P&L and XIRR.

Analytics

- **Accounting:** P&L, weights, FX conversion, time- and money-weighted returns.
- **Risk:** volatility; Sharpe, Sortino, Calmar and Omega ratios; Cornish–Fisher VaR/CVaR with Kupiec backtest; drawdown episodes; PCA, clustering and shrunk covariance; risk attribution; rolling metrics; a 0–100 risk score.
- **Benchmark & attribution:** beta, alpha, tracking error, capture ratios, Fama–French regression with Newey–West errors.
- **Strategy:** efficient frontier, risk parity, Black–Litterman, Monte Carlo, historical stress replay (COVID crash, 2022 rate shock), rebalancing trade lists, Indian STCG/LTCG tax-loss harvesting, covered calls and protective puts (Black–Scholes).
- **Fundamentals & research:** Piotroski F-Score, Altman Z, Beneish M, market news, weekly economic calendar, watchlist price alerts.
- **Export:** formatted Excel workbooks, HTML reports, scheduled email tearsheets.

Machine-Learning Forecaster (v3)

Seven scikit-learn models (RidgeCV, ElasticNetCV, random forest, gradient boosting and baselines) trained on 42 leak-free features using purged walk-forward validation. Skill is measured out-of-sample against a random walk, with Diebold–Mariano significance tests, conformal prediction intervals, a multi-horizon forecast cone, permutation importance and portfolio-level forecasts. When the models fail to beat "price stays flat", the headline says so. Deep learning is deliberately excluded: a few thousand daily price bars cannot support LSTM or transformer models.

Architecture

One rule: **computation never imports Streamlit, presentation never computes, and every network call lives in `src/data/`.**

Layer	Contents
`src/ui/`	17 pages; `pipeline.py` runs a fixed 10-stage analysis (validate → market data → positions → history → risk → benchmark → structure → rolling → ledger → narrative) producing one `AnalysisResult`
`src/analytics/`	23 pure modules, ~217 functions, no I/O
`src/charts/`	51 Plotly chart builders on one theme
`src/data/`	Yahoo Finance, RSS feeds, exchange files, SQLite store

Results are cached per run, and data by how fast it changes: quotes 5 min, history 30 min, fundamentals 24 h. Swapping Yahoo for a broker feed means rewriting one file.

Stack: Python 3.11+, pandas, numpy, scipy, scikit-learn, yfinance, Plotly, Streamlit 1.50+, SQLite, XlsxWriter, pytest; Docker, Makefile and GitHub Actions CI.

Testing: ~4,544 offline, seeded tests checked against hand-computed values, at 92% coverage (per the README). No browser, network or market hours are needed.

Limitations

- Yahoo prices are ~15 minutes delayed; the app labels them as such.
- Without a ledger, history assumes today's share counts throughout.
- The optimiser is single-period, in-sample mean-variance; stress betas are first-order CAPM; benchmarks exclude dividends.
- The forecaster sees price only, usually reports no skill, and is not a cost-aware backtest.
- Yahoo publishes no NSE/BSE option chains; fundamentals are restated, not point-in-time.

Roadmap

Real-time Zerodha Kite/Upstox feed, cross-sectional universe-wide forecasting, a cost-aware strategy backtest including STT, market-regime detection, and point-in-time fundamentals as forecaster features.